



# Post Hoc Analyses

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## 1. What are Post Hoc Analyses? What is Anova testing?

POST Hoc Analysis refers to the various statistical methods used after ANOVA testing is performed to determine the specific pairwise differences between the groups. While ANOVA testing can identify that there is a difference among all tested groups, we need post hoc analysis to detect where those differences are.

Power can be thought of as how often a post hoc test will find these differences, though this is regardless of whether or not there actually is a difference. Low power is less likely to claim there is a difference, while high power is more likely to claim there is a difference. Our discussion will focus on various post hoc tests, their power, strengths, and weaknesses.

## 2. Which of These Post Hoc Tests Will we Examine?

| Test                  | Power | Type         | Best Use Case                  |
|-----------------------|-------|--------------|--------------------------------|
| Bonferroni Correction | Low   | Conservative | Teaching, simple use.          |
| Tukey's HSD           | Low   | Conservative | Equal variances, all-pairs.    |
| Scheffé's Method      | Low   | Conservative | A vs B,C,D.                    |
| Fisher's LSD          | High  | Liberal      | Small experiments.             |
| Holm-Bonferroni       | Mid   | Conservative | More power than Bonferroni.    |
| Dunnett's Test        | Mid   | Conservative | Control vs. experimental.      |
| Benjamini-Hochberg    | High  | Liberal      | Many comparisons, FDR control. |

Table 1: Summary of Post Hoc Tests, Power, and Best Use Cases.

## 3. Simulating our Data and Gathering Statistics

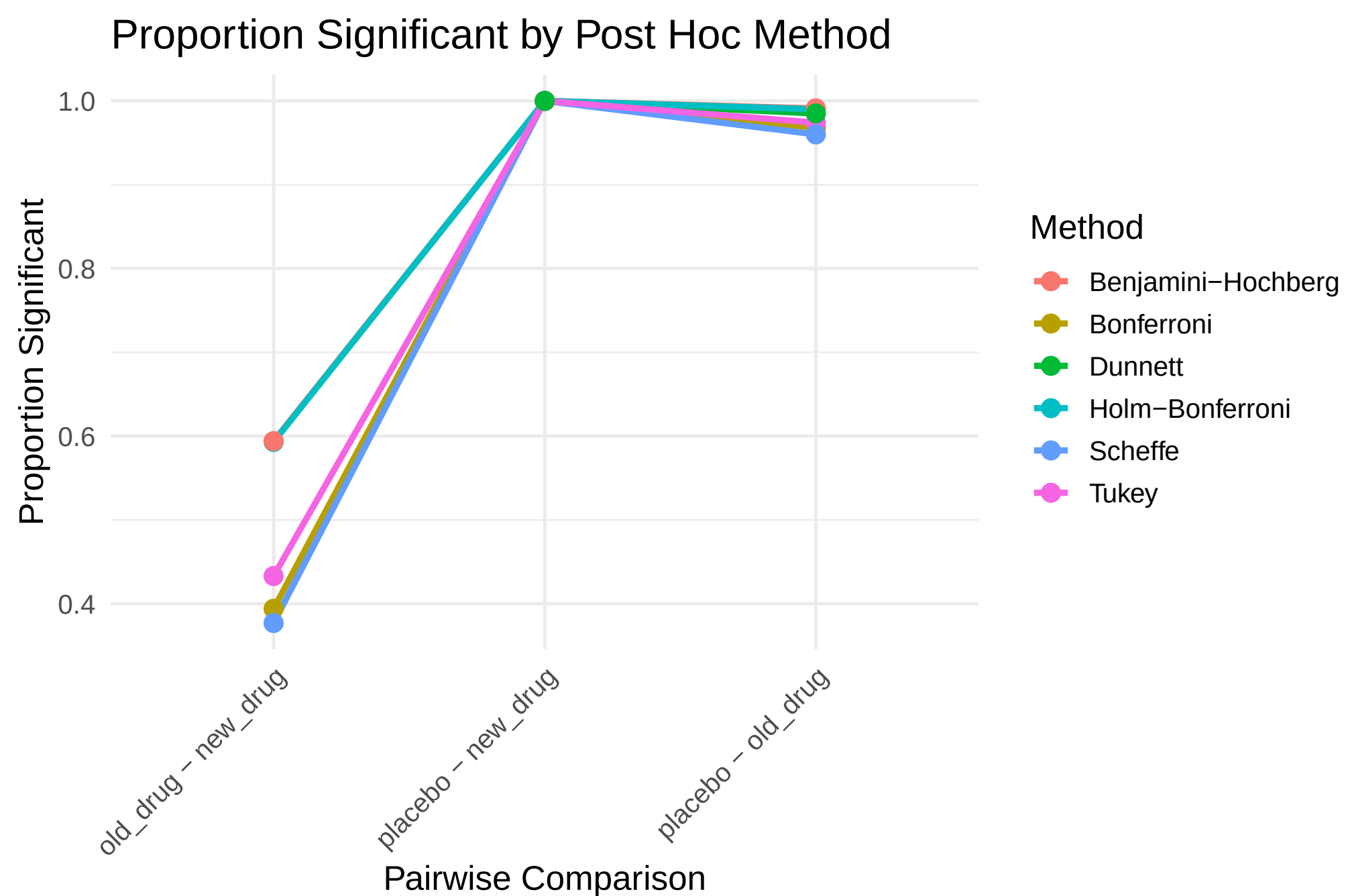
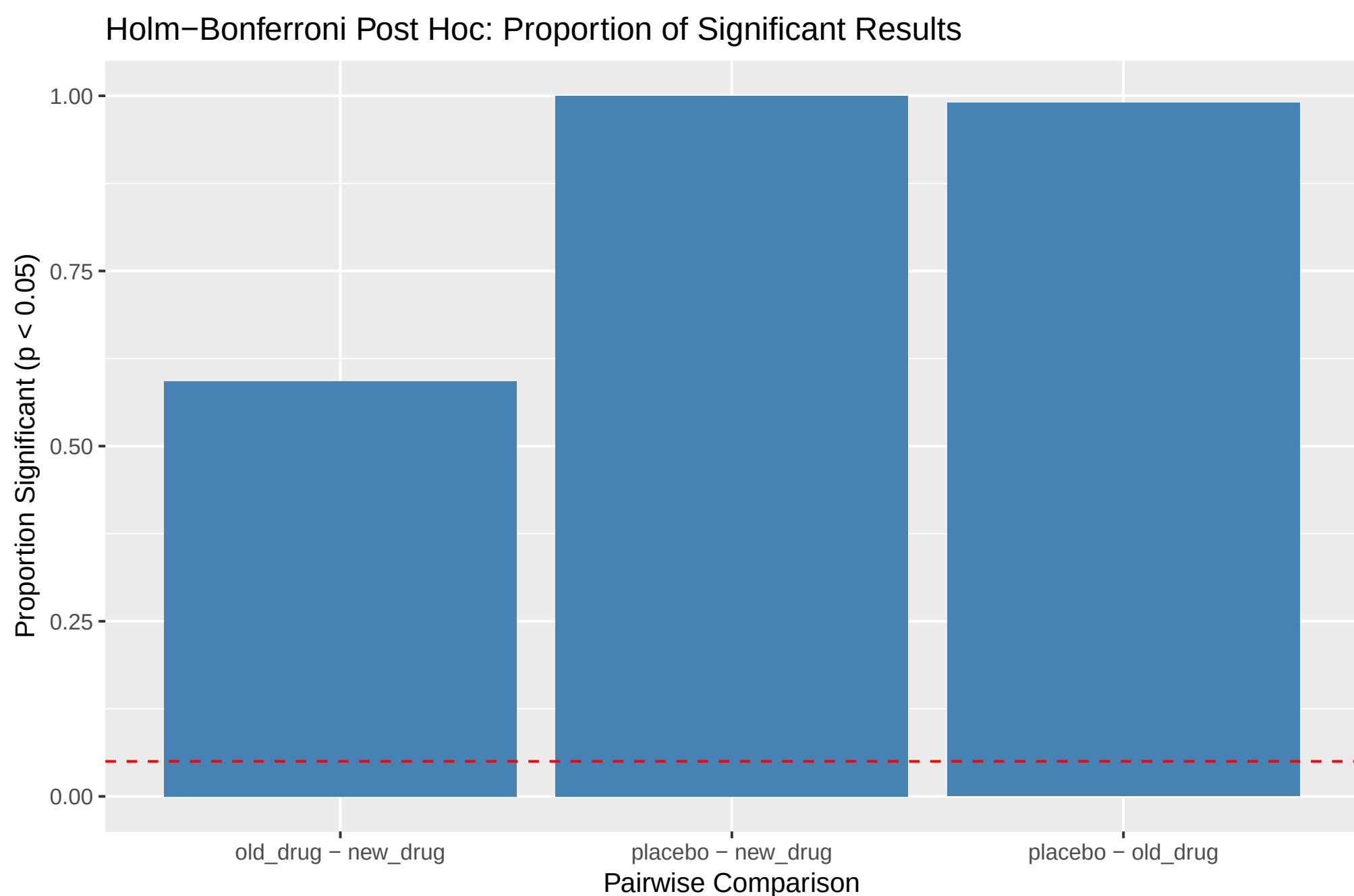
1. Use rnorm to generate 3 different normally distributed data sets, with same sample size and standard deviation, but different means.
2. Write a function to generate 1000 simulations of each data set (same parameters but different data points)
3. Run ANOVA on every triad, which will determine if there is a difference between their means
4. Follow up ANOVA with a Post Hoc Test, to determine where those differences lie
5. Repeat until all Post Hoc Tests are exhausted on all 1000 pairwise comparisons

## 4. Illustrating our Findings

FOR illustrative purposes, much of the output for one of our post hoc tests will be included below, so that it may be an example largely representative of how the other tests look in terms of output.

| Comparison          | Prop Significant | Mean P-value |
|---------------------|------------------|--------------|
| Old Drug - New Drug | 0.593            | 0.1146882    |
| Placebo - New Drug  | 1.000            | 0.0000642    |
| Placebo - Old Drug  | 0.990            | 0.00326      |

Table 2: Holm-Bonferroni Results Summary



An effective way to interpret the results of the graph are to imagine the gray space above the line representing failure of the test. We generated the data and made sure there were real differences in the mean, so the only error our tests can make is not detecting that difference. This is called a type II error, when a difference is really present, and our test fails to detect it.

## 5. Applications in Real Studies

IN Holm-Bonferroni, we can observe through the graph that it only majorly failed to detect the difference between the old drug and the new drug, which was the smallest difference, theoretically the hardest to detect. Through an understanding of the mechanics of the Holm-Bonferroni test, we know that it is a generally conservative test. Again, this means it will generally protect more against Type I errors, or finding a difference when there isn't one. This is especially valuable in drug studies, as ethically it is more detrimental to green-light a drug that is either ineffective, or not as effective as advertised, than it is to take longer in development to make sure that the product is actually an advancement.

Still, companies want to make developments sooner rather than later, and spend less money in the research and development phase. These motivators conflict, and the only way out is to find a balance.

## 6. Conclusion

IN Holm-Bonferroni, we find a test that well represents that desired balance in our experiment. Even though it is largely known as a conservative test, it was able to detect the largest difference 100 percent of the time, the mid-dling difference 99 percent of the time, and even able to detect our largest difference a majority of the time. This implies that the test should well protect against false positives, while still being considerably powerful. As such, it stands out as the most obvious choice should we decide to move forward and start gathering real data for real advancements.

## References

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